Shane Khaykham Dang

shanedang17@gmail.com | (509) 717-8600 | linkedin.com/in/shane-dang/ | Tacoma, WA

Summary

Results-driven Data Engineer with 3+ years of experience designing and maintaining scalable ETL pipelines, data warehouses, and analytics infrastructure. Proven track record reducing data processing time and query runtimes by up to 60% and 40% respectively. Skilled in Python, SQL, PostgreSQL, and cloud-adjacent data tooling with a strong foundation in data modeling, API integration, and geospatial analysis. Thrives in cross-functional environments translating raw data into reliable, self-serve systems for research and operational teams.

Experience

Systems Analyst, Programmer | TRAC-UW | Seattle, WA **Mar 2024 - Present**

- Co-led a GIS training course for tribal employees across Washington state teaching ArcGIS Pro workflows to produce strip maps compliant with BIA and NTTFI federal reporting standards; also trained participants in Road Safety Audit (RSA) methodology through a hands-on mock RSA on local roadways.
- Designed and maintained production ETL pipelines in Python and SQL processing millions of rows across 4 transportation datasets, reducing manual data preparation time by 60% and shortening analysis turnaround for researchers.
- Engineered a PostgreSQL data warehouse consolidating fragmented transportation datasets across multiple research initiatives; implemented star schema design, indexing strategies, and query optimization that reduced analytical query runtimes by 40%.
- Developed automated data ingestion and validation pipelines in Python (Pandas, NumPy) to extract, clean, and load data from heterogeneous sources including APIs, flat files, and legacy databases improving pipeline reliability and reducing downstream reporting errors.
- Integrated multiple internal and external data sources through REST APIs and automation scripts, enabling unified cross-dataset analysis and eliminating data silos across research teams.
- Built and deployed interactive data dashboards and reporting tools using JavaScript and D3.js, replacing static monthly reports with live, self-serve analytics for researchers and transportation agency stakeholders.
- Architected and maintained on-premises research data infrastructure (Proxmox, TrueNAS) ensuring 99%+ uptime for multi-terabyte transportation datasets; recovered critical datasets from a storage failure achieving 95% data recovery.

GIS Data Analyst | TRAC-UW | Seattle, WA **Jul 2023 - Mar 2024**

- Diagnosed and resolved performance bottlenecks in data processing pipelines and visualization tools, achieving a 30% reduction in system downtime and a 25% improvement in performance across tools serving 5,000+ users.
- Integrated sidewalk and walkway datasets from multiple government and survey sources; applied data cleaning and standardization techniques to build a unified accessibility layer used in regional planning tools, resulting in a 20% increase in accessibility ratings.
- Conducted geospatial analysis of transportation infrastructure using SQL, spatial joins, and clustering techniques to improve classification accuracy of regional sidewalk network data.
- Refactored and optimized geospatial data processing scripts to automate repetitive analysis workflows, reducing manual intervention and accelerating delivery of transportation research datasets.

Awards

Richard Thomas Award - 1st Place | WAGISA DTA Competition, Tacoma, WA **2023**

Education

B.A. Geography (Data Science Focus) | University of Washington | Seattle, WA **2018 - 2023**

Skills

- **Languages:**
  Python (Pandas, NumPy, SQLAlchemy), SQL, JavaScript, R, Java, HTML, CSS, PHP
- **Data Engineering:**
  ETL Pipeline Development, Data Modeling, Star Schema Design, Data Warehousing, Query Optimization, Data Architecture, Batch & Incremental Processing
- **Databases:**
  PostgreSQL, MySQL, SQL Server; familiar with BigQuery and Snowflake concepts
- **Cloud & Platforms:**
  REST API Integration, Git/GitHub, Linux/Bash, Proxmox, TrueNAS, Visual Studio Code, DBeaver
- **Data Analysis & Automation:**
  Data Cleaning & Preprocessing, API Integration, Automation Scripting, Time Series Analysis, Exploratory Data Analysis (EDA), Feature Engineering
- **Data Visualization:**
  D3.js, Altair, Matplotlib, Interactive Dashboards, ArcGIS, QGIS, Spatial Data Analysis